

Q1 Grade Calculator *(25 marks)*

Problem Statement

Write a C program to input marks (0-100) of a student and display the grade using the following conditions:

- A: 90-100
- B: 80-89
- C: 70-79
- D: 60-69
- F: Below 60

Code

```
#include <stdio.h>
int main(){
int mark;
scanf("%d", &mark);

if(mark<0 || mark>100){
printf("invalid mark entered");
}
else if(mark>=90){
printf("grade a");
}
else if(mark>=80){
printf("grade b");
}
else if(mark>=70){
printf("grade c");
}
else if(mark>=60){
printf("grade d");
}
```

```
else{  
printf("grade f");  
}  
return 0;  
}
```

Sample Input

3

Sample Output

grade f

Q2 Largest of Three Numbers *(25 marks)*

Problem Statement

Write a C program to input three integers and determine the largest number using nested if-else statements.

Code

```
#include <stdio.h>
int main(){
int a, b, c;
scanf("%d", &a, &b, &c);
if(a>b){
    if(a>c)
        printf("a is greater");
    else
        printf("c is greater");
}
else{
    if(b>c)
        printf("b is greater");
    else
        printf("c is greater");
}
return 0;
}
```

Sample Input

```
122
32
13
```

Sample Output

```
a is greater
```

Q3 Inverted Right-Angled Triangle *(25 marks)*

Problem Statement

Write a C program to print the following pattern.

- *****
- ****
- ***
- **
- *

Code

```
#include <stdio.h>
int main(){
    int i,j;
    for(i=5;i>=1;i--){
        for (j=1;j<=i;j++){
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Sample Input

stdin input (none required)

Sample Output

```
*****
****
***
**
*
```

Q4 Prime Numbers in a Range *(25 marks)*

Problem Statement

Write a C program to input two integers start and end and print all prime numbers between them using nested loops and appropriate control statements.

Code

```
#include <stdio.h>
int main(){
int start, end, i, j, flag;
scanf("%d %d", &start, &end);
for (i=start; i<=end; i++){
if(i<2)
continue;
flag=1;
for (i=2;j<=i/2;j++){
if(i%j==0){

}
}
}
}
```

Sample Input

10 30

Sample Output

(No output is produced - the program compiles but never reaches a printf statement, and the inner loop reuses "i" as its own counter and never initializes "j", so the prime check never runs correctly.)